



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

17 September 2018

2 hours

Candidates answer on the Question Paper.

No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **ONE** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 29
2	/ 13
3	/ 8
Section B	
4 or 5	/ 25
Total	/ 75

Section A

Answer **all** the questions in this section.

QUESTION 1

DNA analysis revealed that 47 young children buried more than 1500 years ago near Rome in Italy died during an epidemic of an infectious disease. The nucleotide sequence of the 18S rRNA gene of *Plasmodium falciparum* was positively identified in some of the bones. *P. falciparum* is a parasite that causes the disease malaria.

- (a) (i) Describe the process by which repeated cycles of PCR could specifically amplify the *P. falciparum* 18S rRNA gene nucleotide sequences from a small quantity of the DNA from the buried bones. [4]

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- (ii) Suggest why the 18S rRNA gene sequence is used for the positive identification of *P. falciparum*. [2]

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P. falciparum infects a person when a mosquito carrying the parasite injects an anti-coagulant into the bloodstream prior to feeding on blood. The parasites multiply in liver cells before emerging after 9-30 days wrapped in liver cell surface membrane. They enter red blood cells, multiply and then cause rupture of the host cells and release of more parasites every 36-48 hours, in a manner that has some similarity to that of viruses.

- (b) (i)** The symptoms of malaria occur repeatedly every 36-48 hours and include body pain. These symptoms are associated with many other infectious diseases, including those caused by the viruses.

Suggest how *P. falciparum* may cause these types of symptoms. [2]

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- (ii)** Use the information given and your own knowledge, suggest why it is extremely difficult to design an effective vaccine against malaria. [3]

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- (c) Malaria was common in Italy, a European country situated in the Northern Hemisphere. Widespread land drainage together with the use of the insecticide DDT and the drug chloroquine eradicated both the mosquito vectors and the malarial parasites in the 1950s. Due to the success of these measures, they were later discontinued as they were no longer necessary.

Articles in the scientific literature more recently show that malarial mosquitoes are returning to Italy and increasing their numbers and their northerly range, with some cases of malaria being reported. In general, winters are milder and summers are hotter in the south of the country, with temperatures decreasing to the north, especially in the winters.

Discuss whether the return of malaria to Italy can be attributed to climate change. [4]

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- (d) Chloroquine-resistant (CQR) malarial parasites, *P. falciparum*, were first reported in 1950s and are now widespread. The resistance is caused by mutations of a gene known as *pfcr*.

- (i) Explain whether resistance to chloroquine in *P. falciparum* is an example of continuous or discontinuous variation. [2]

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(ii) Explain why CQR *P. falciparum* are now widespread. [4]

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Analysis of *pfcr*t in CQR *P. falciparum* from different parts of the world shows differences in one section of the gene only. The amino acid sequences coded for by this section are shown in Table 1.1, together with the amino acid sequence coded for by the allele found in parasites that are still susceptible to chloroquine (non-CQR). The amino acid sequence coded for by the rest of the gene is the same in all cases.

Table 1.1

allele of <i>pfcr</i> t gene	amino acid sequence coded for by allele of <i>pfcr</i> t gene
non-CQR	—Cys—Met—Asn—Lys—His—Ala—Glu—Asn—Ile —Met—
CQR from Africa	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Ile —Ile —
CQR from Asia	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser —Thr—Ile —
CQR from Peru and Brazil	—Ser—Met—Asn—Thr—His—Ser—Gln—Asp—Leu—Arg—
CQR from Colombia	—Cys—Met—Glu—Thr—Gln—Ser—Gln—Asn—Ile —Thr—

(iii) With reference to Table 1.1, calculate the mean number of different amino acids coded for by CQR alleles in comparison with the non-CQR allele. Show your working. [2]

mean number of different amino acids

In the non-CQR *P. falciparum*, chloroquine accumulates in the digestive vacuole of *P. falciparum* and interferes with the detoxification of haem in the host red blood cell. This results in parasite death.

Research shows that there is a point mutation in the *pfcr* gene of CQR *P. falciparum* which leads to the formation of a chloroquine-resistance transporter located in the digestive vacuole membrane of CQR *P. falciparum*.

This mutation changes the amino acid from lysine to threonine at the binding site of the chloroquine-resistance transporter.

(e) Using the information provided, suggest how this mutation of *pfcr* gene increases the chloroquine resistance in *P. falciparum*. [3]

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In 2014, the World Health Organisation (WHO) estimated that 3200 million people were at risk of malaria. This was almost half of the world population in 2014.

Table 1.2 shows the number of cases of malaria and the number of deaths from malaria between 1998 and 2013. The table shows numbers for all countries of the world and for the countries in the WHO African region.

The table also shows the numbers in the African region as percentages of the numbers for all countries.

Table 1.2

year	number of cases of malaria in millions		cases in the African region as a percentage of all countries/ %	number of deaths from malaria in thousands		deaths in the African region as a percentage of all countries/ %
	all countries	African region		all countries	African region	
1998	272.9	237.6	87.1	1067.0	961.0	90.1
2003	236.0	186.6	79.1	872.0	800.0	91.7
2008	225.1	181.0	80.4	747.0	677.0	90.6
2013	198.0	158.4	80.0	584.0	525.6	90.0

(f) Describe the trends shown in Table 1.2.

[3]

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[Total: 29]

QUESTION 2

Some mothers of infants with Down syndrome have a mutation of a gene coding for an enzyme involved in the metabolism of methyl groups. A lack of methyl groups on the centromeric DNA results in mistakes in the segregation of chromosomes during cell division.

- (a) Suggest how a lack of methyl groups on the centromeric DNA results in mistakes in segregation of chromosomes during cell division. [2]

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The gene mutation referred to in (a) also affects the metabolism of the vitamin folic acid. Abnormal folic acid metabolism is seen in mothers of infants with neural tube disorders (NTD).

Two groups of families, in Israel and Ukraine, were investigated to see whether there was any link between the incidence of Down syndrome and NTD. The results are shown in Table 2.1.

Table 2.1

group of families	number of pregnancies	number of cases of Down syndrome	expected number of cases of Down syndrome	probability of result occurring by chance
families at risk of NTD (Israel)	1492	11	1.87	$p = 0.00001$
group of families	number of pregnancies	number of cases of NTD	expected number of cases of NTD	probability of result occurring by chance
families at risk of Down syndrome (Ukraine)	1847	7	1.37	$p = 0.001$

- (b) With reference to Table 2.1, justify the claim that there is a link between Down syndrome and NTD in these families. [3]

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(c) NTD is due to recessive mutation on a gene found on chromosome 9.

Complete Table 2.2 to show the results of the calculations.

[2]

Table 2.2

parent 1	parent 2	percentage probability of having a child with NTD	percentage probability of having an unaffected child (not carrier)	percentage probability of having a child who is a carrier
unaffected	carrier	0	50	50
carrier	carrier			
unaffected	has NTD			
carrier	has NTD	50	0	50

(d) Scientists investigated the use of induced pluripotent stem cells (iPS cells) to treat NTD in mice. The procedures are as follows:

- The skin of mice with NTD were collected within the first few hours of birth
- The skin-derived fibroblast cultures were established
- Plasmids containing four transcription factors to reprogramme skin cells to form iPS cells were introduced into the fibroblast cultures.
- The scientist then stimulated the *in vitro* differentiation of iPS cells into normal neural cells.
- The normal neural cells were transplanted back to mice.

Explain how transcription factors can reprogramme cells to form iPS cells.

[2]

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- (e)** The iPS cells were first discovered in 2006 and have been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine.

Recent research shows that abnormal reprogramming can occur during the induction of human induced pluripotent stem cells (HiPS cells) and tumours could be generated.

- (i)** Compare the embryonic stem cell and HiPS cell. [2]

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- (ii)** Comment on the ethical issues involved in the use of HiPS cells. [2]

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[Total: 13]

QUESTION 3

Immediately following an infection with human immunodeficiency virus (HIV), there is a period of high levels of HIV in the blood (viremia). HIV is an intracellular pathogen and would stimulate a strong CD8 cytotoxic T cell response. Antibodies against HIV glycoproteins and nucleocapsid proteins are also produced. The immune response against HIV results in a reduction in viremia but does not eliminate the virus completely.

- (a) It is known that HIV infects CD4 T helper cells. Suggest other features about HIV that make it difficult for the immune response to eliminate it. [3]

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Fig. 3.1 shows a test that has been developed to find out if a person has antibodies to the HIV antigen.

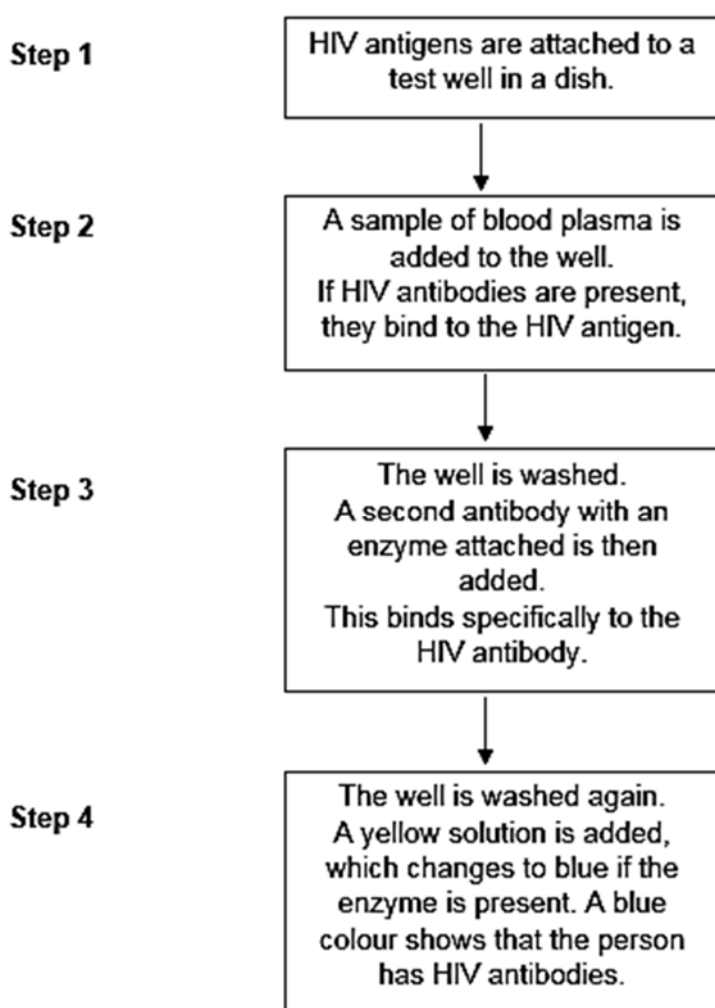


Fig. 3.1

(b) This test only detects the presence of HIV antibodies. Suggest why it cannot be used to find out if a person has AIDS. [1]

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(c) Explain why the solution will remain yellow if a person is **not** infected with HIV. [2]

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(d) A mother who was infected with HIV gave birth to a baby. The baby was tested positive using this test. Suggest why this does not prove that the baby is infected with HIV. [2]

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[Total: 8]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 4

- (a)** Evolutionary studies suggest that some membrane-bound organelles in plant cells have evolved from a symbiotic relationship between eukaryotic and prokaryotic cells. Molecular phylogenetic techniques revealed similarities between the structures of the organelles and that of the prokaryotic cells.

Describe how structures involved in photosynthesis provide evidence for endosymbiosis and outline how molecular techniques such as gel electrophoresis can be used to show this phylogenetic relationship between prokaryotes and eukaryotes. [10]

- (b)** Climate change has been attributed to a rise in carbon dioxide levels, which may affect organisms directly or indirectly.

Explain how an increase in carbon dioxide levels may affect plants directly and indirectly. [15]

[Total: 25]

QUESTION 5

- (a)** All cells, from bacteria to humans, express their genetic information via the central dogma of molecular biology which depicts the flow of genetic information from DNA to protein.

Outline the processes involved in the central dogma of molecular biology, and explain, using specific examples, why the central dogma may not always hold true. [10]

- (b)** The expression of genes gives rise to functional proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of proteins in the control of responses in organisms. [15]

[Total: 25]

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[illegible]

[illegible]

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[illegible]

This image shows a full page of a document template designed for handwriting practice or general note-taking. It consists of approximately 28 evenly spaced horizontal dotted lines across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

End of Paper